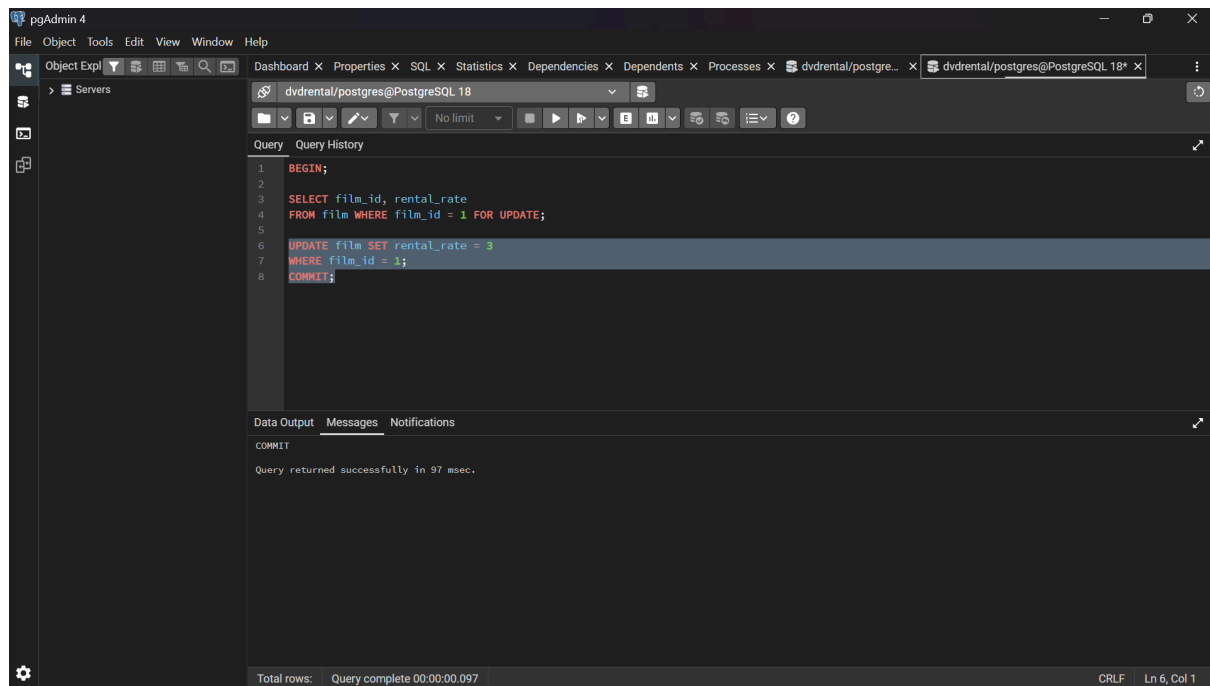


Lab 1:



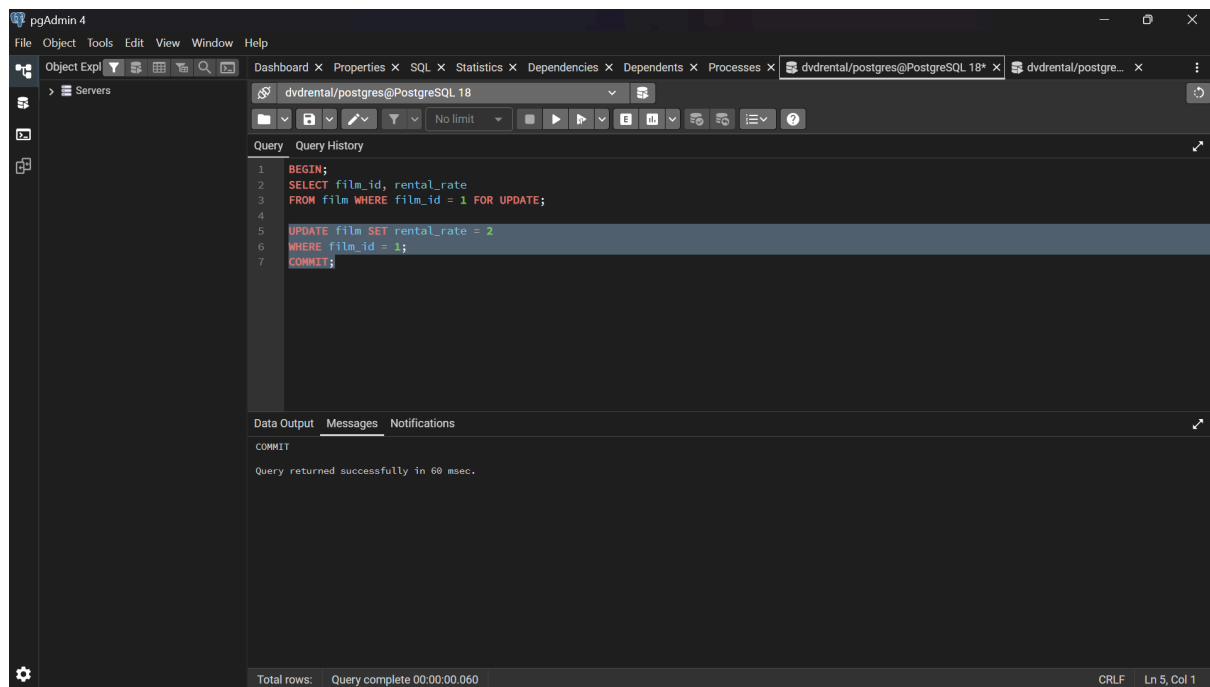
A screenshot of the pgAdmin 4 interface. The top menu bar includes File, Object, Tools, Edit, View, Window, and Help. The left sidebar shows the 'Servers' tree. The main pane displays a SQL query in the 'Query' tab, with the 'Query History' tab also visible. The query is as follows:

```
1 BEGIN;  
2  
3 SELECT film_id, rental_rate  
4 FROM film WHERE film_id = 1 FOR UPDATE;  
5  
6 UPDATE film SET rental_rate = 3  
7 WHERE film_id = 1;  
8 COMMIT;
```

The 'Data Output' tab shows the result of the query:

```
COMMIT  
  
Query returned successfully in 97 msec.
```

The status bar at the bottom indicates 'Total rows: Query complete 00:00:00.097' and 'CRLF Ln 6, Col 1'.



A screenshot of the pgAdmin 4 interface, similar to the one above. The main pane displays a SQL query in the 'Query' tab, with the 'Query History' tab also visible. The query is as follows:

```
1 BEGIN;  
2 SELECT film_id, rental_rate  
3 FROM film WHERE film_id = 1 FOR UPDATE;  
4  
5 UPDATE film SET rental_rate = 2  
6 WHERE film_id = 1;  
7 COMMIT;
```

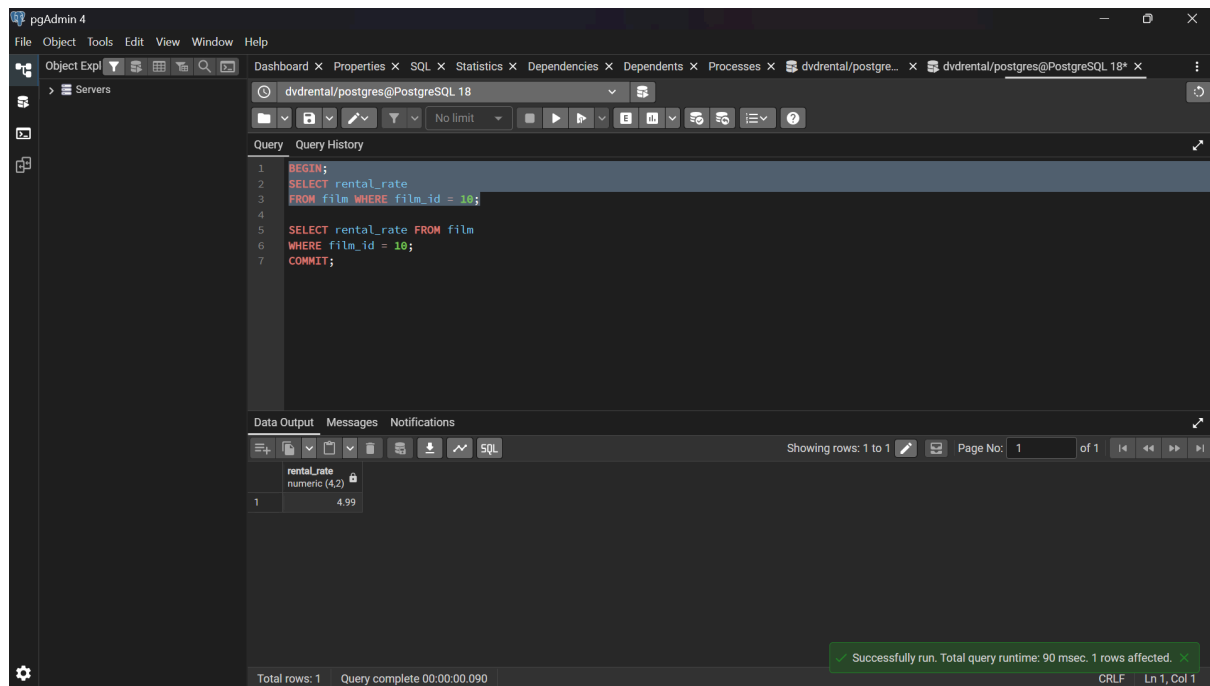
The 'Data Output' tab shows the result of the query:

```
COMMIT  
  
Query returned successfully in 68 msec.
```

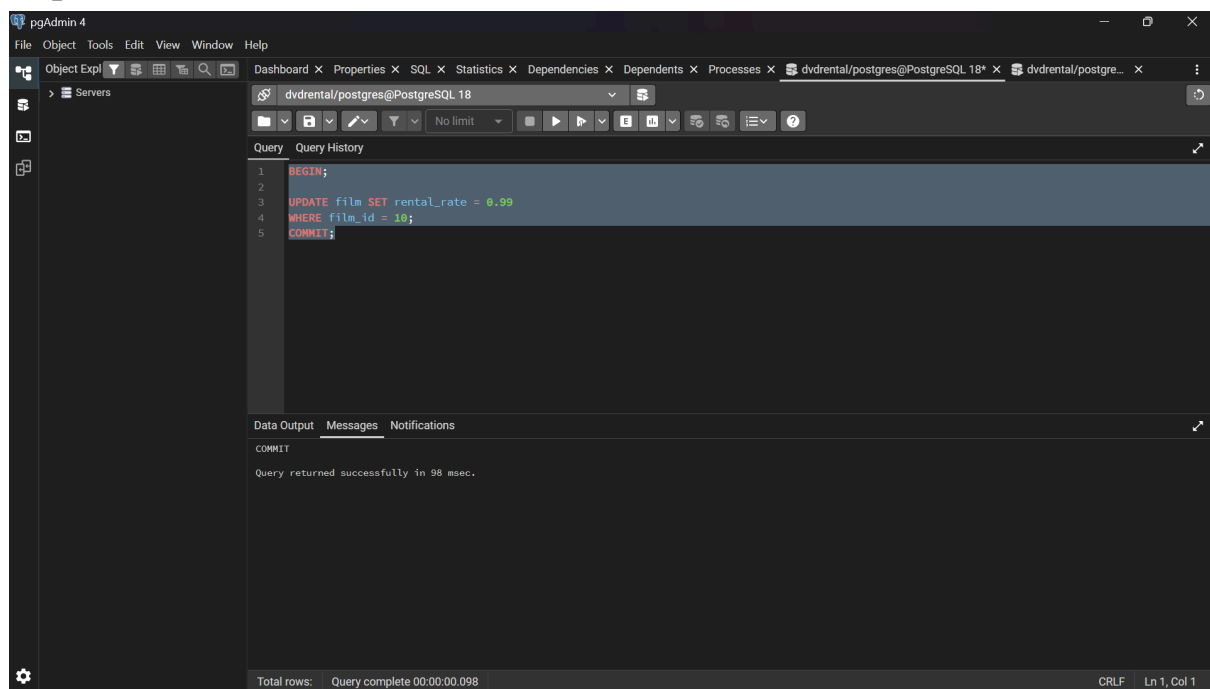
The status bar at the bottom indicates 'Total rows: Query complete 00:00:00.060' and 'CRLF Ln 5, Col 1'.

Lab 2:

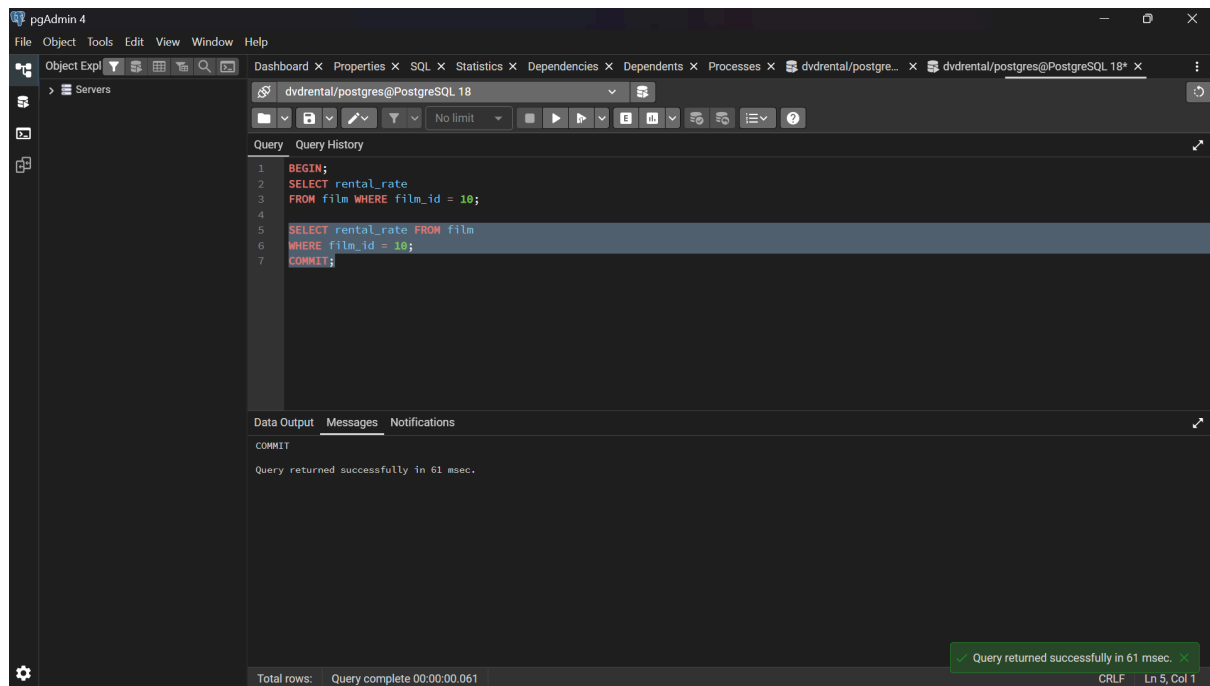
step 1:



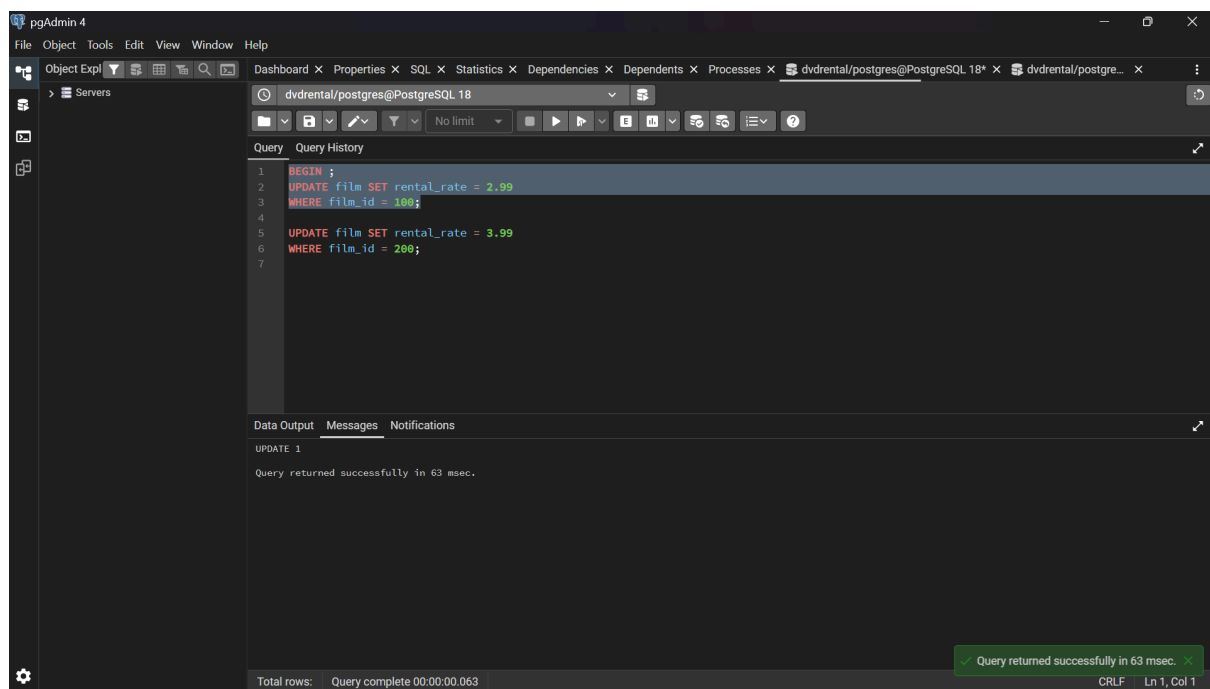
step 2:

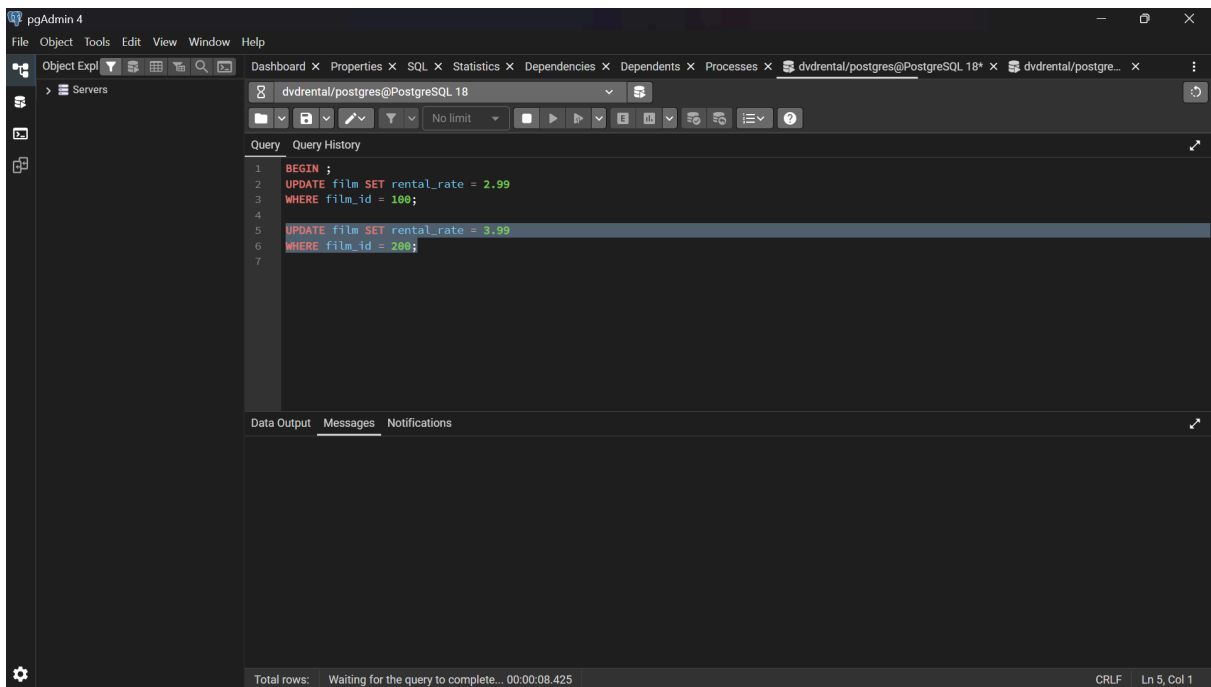
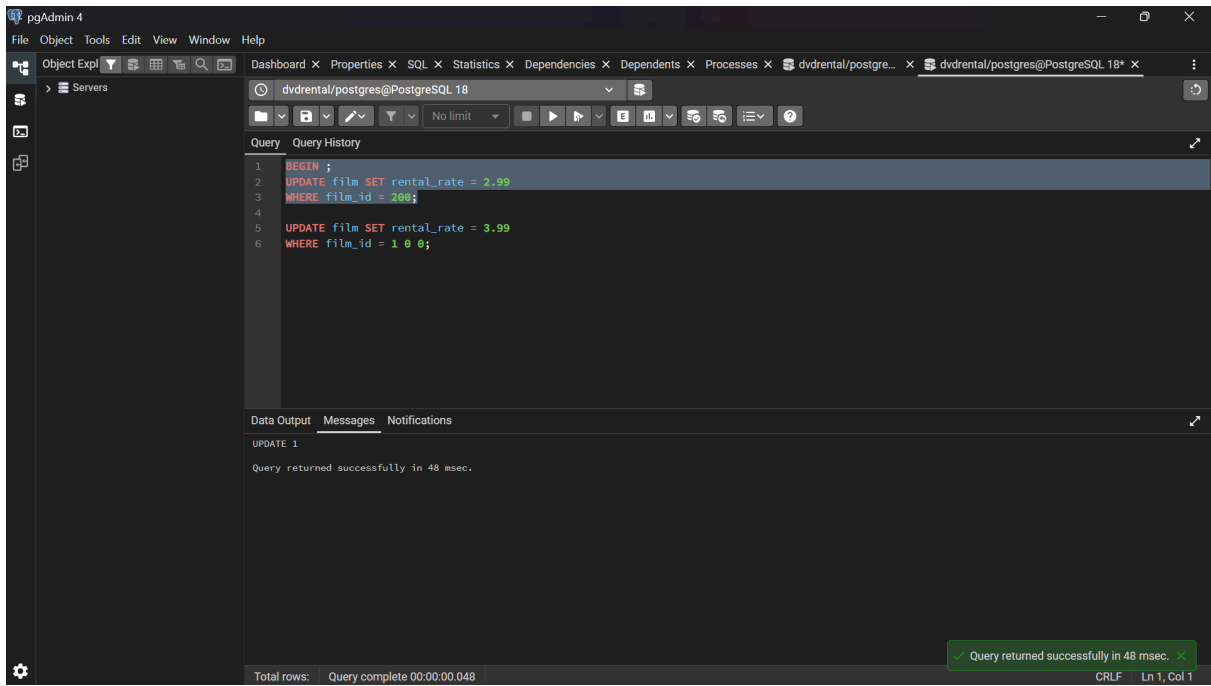


step 3:



Lab 3:





The screenshot shows the pgAdmin 4 interface with a SQL query window. The query is a transaction block that updates rental rates for two different film IDs. The first update for film_id 200 is successful, but the second update for film_id 100 fails due to a deadlock (SQL state: 25P02).

```
1 BEGIN ;
2 UPDATE film SET rental_rate = 2.99
3 WHERE film_id = 200;
4
5 UPDATE film SET rental_rate = 3.99
6 WHERE film_id = 100;
```

Data Output Messages Notifications

ERROR: current transaction is aborted, commands ignored until end of transaction block

SQL state: 25P02

Total rows: Query complete 00:00:00.087 CRLF Ln 5, Col 1

The screenshot shows the pgAdmin 4 interface with a SQL query window. The query is a transaction block that updates rental rates for two different film IDs. The first update for film_id 100 is successful, and the second update for film_id 200 is also successful. The transaction is completed successfully.

```
1 BEGIN ;
2 UPDATE film SET rental_rate = 2.99
3 WHERE film_id = 100;
4
5 UPDATE film SET rental_rate = 3.99
6 WHERE film_id = 200;
```

Data Output Messages Notifications

UPDATE 1

Query returned successfully in 22 secs 441 msec.

Total rows: Query complete 00:00:22.441 CRLF Ln 5, Col 1

Lab 4:

pgAdmin 4

File Object Tools Edit View Window Help

Dashboard × Properties × SQL × Statistics × Dependencies × Dependents × Processes × dvdrental/postgres@PostgreSQL 18* × dvdrental/postgre...

Servers

Query Query History

```
1 BEGIN;
2 SELECT * FROM inventory
3 WHERE inventory_id = 10 FOR UPDATE;
4
5 INSERT INTO rental
6 (rental_date, inventory_id, customer_id, staff_id)
7 VALUE (NOW(), 10, 1, 1);
8 COMMIT;
```

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

inventory_id [PK] integer	film_id smallint	store_id smallint	last_update timestamp without time zone
1	10	2	2006-02-15 10:09:17

Total rows: 1 Query complete 00:00:00.117

Successfully run, Total query runtime: 117 msec, 1 rows affected.

pgAdmin 4

File Object Tools Edit View Window Help

Dashboard × Properties × SQL × Statistics × Dependencies × Dependents × Processes × dvdrental/postgre... × dvdrental/postgres@PostgreSQL 18* ×

Servers

Query Query History

```
1 BEGIN;
2 SELECT * FROM inventory
3 WHERE inventory_id = 10 FOR UPDATE;
4
5 ROLLBACK;
```

Data Output Messages Notifications

ERROR: current transaction is aborted, commands ignored until end of transaction block

SQL state: 25P02

Total rows: Query complete 00:00:00.053

pgAdmin 4

File Object Tools Edit View Window Help

Dashboard × Properties × SQL × Statistics × Dependencies × Dependents × Processes × dvdrental/postgres@PostgreSQL 18* × dvdrental/postgre...

Servers

Query Query History

```
1 BEGIN;
2 SELECT * FROM inventory
3 WHERE inventory_id = 10 FOR UPDATE;
4
5 INSERT INTO rental
6 (rental_date, inventory_id, customer_id, staff_id)
7 VALUES (NOW(), 10, 1, 1);
8 COMMIT;
```

Data Output Messages Notifications

ERROR: current transaction is aborted, commands ignored until end of transaction block

SQL state: 25P02

Total rows: 1 Query complete 00:00:00.100 CRLF Ln 5, Col 1

pgAdmin 4

File Object Tools Edit View Window Help

Dashboard × Properties × SQL × Statistics × Dependencies × Dependents × Processes × dvdrental/postgre... × dvdrental/postgres@PostgreSQL 18* ×

Servers

Query Query History

```
1 BEGIN;
2 SELECT * FROM inventory
3 WHERE inventory_id = 10 FOR UPDATE;
4
5 ROLLBACK;
```

Data Output Messages Notifications

ROLLBACK

Query returned successfully in 40 msec.

Total rows: Query complete 00:00:00.040 ✓ Query returned successfully in 40 msec. × CRLF Ln 5, Col 1